intervention applied at one layer must be accounted for before introducing an intervention at the subsequent layer. To capture this causal structure, *Mean Activation Transport (Mean-AcT)* in (Rodriguez et al., 2025) estimates the steering vectors incrementally at each layer as follows:

$$\boldsymbol{x}_i(k-1,\boldsymbol{q}) = \rho_{\text{steer}}(\boldsymbol{x}_i(k-1,\boldsymbol{q}),\boldsymbol{r}(k-1)), \text{for } \boldsymbol{q} \in \mathcal{D}_{\text{source}} \tag{4}$$

$$\boldsymbol{x}_i(k,\boldsymbol{q}) = f_i^{(k)}(\boldsymbol{x}(k-1,\boldsymbol{q})), \text{for } \boldsymbol{q} \in \mathcal{D}_{\text{source}} \cup \mathcal{D}_{\text{target}} \tag{5}$$

$$\boldsymbol{\mu}_{\text{target}}(k) = \frac{1}{|\mathcal{D}_{\text{target}}^{(\text{train})}|} \sum_{i \in I, \boldsymbol{q} \in \mathcal{D}_{\text{target}}^{(\text{train})}} \boldsymbol{x}_i(k,\boldsymbol{q}), \qquad \boldsymbol{\mu}_{\text{source}}(k) = \frac{1}{|\mathcal{D}_{\text{source}}^{(\text{train})}|} \sum_{i \in I, \boldsymbol{q} \in \mathcal{D}_{\text{source}}^{(\text{train})}} \boldsymbol{x}_i(k,\boldsymbol{q})$$

$$\boldsymbol{r}(k) = \boldsymbol{\mu}_{\text{target}}(k) - \boldsymbol{\mu}_{\text{source}}(k). \tag{6}$$

Like ActAdd, Mean-AcT sets $\rho_{\text{steer}}(\boldsymbol{x}(k),\boldsymbol{r}(k)) = \boldsymbol{x}(k) + \alpha \boldsymbol{r}(k)$.

2.3 Proportional–Integral–Derivative Controller

Proportional-Integral-Derivative (PID) control is a feedback mechanism extensively used in control systems (Minorsky, 1922). It is valued for its simplicity, robustness, and effectiveness in a broad range of applications, from industrial automation to robotics and aerospace systems (Visioli, 2006; Borase et al., 2021). The core idea behind PID control is to compute a control signal based on the error between a target reference signal and the actual output of a system. Specifically, consider a continuous-time dynamical system governed by a state space model

$$\dot{\boldsymbol{x}}(t) = g(\boldsymbol{x}(t),\boldsymbol{u}(t),t), \boldsymbol{y}(t) = h(\boldsymbol{x}(t),\boldsymbol{u}(t),t), \tag{7}$$

where $\boldsymbol{x}(t) \in \mathbb{R}^d$ denotes the state variable, $\boldsymbol{u}(t) \in \mathbb{R}^m$ is the control variable, and $\boldsymbol{y}(t) \in \mathbb{R}^{d'}$ represents the measured output signal. Here, $g : \mathbb{R}^d \times \mathbb{R}^m \to \mathbb{R}^d$ specifies the system dynamics, and $h : \mathbb{R}^d \times \mathbb{R}^m \to \mathbb{R}^{d'}$ is an output mapping. A PID controller applies the control variable $\boldsymbol{u}(t)$ to minimize the discrepancy between a target reference, or also known as the setpoint in the literature of PID control, $\boldsymbol{y}_{sp}(t)$ and the actual output $\boldsymbol{y}(t)$. This discrepancy, called control error, is defined as

$$\boldsymbol{e}(t) = \boldsymbol{y}_{sp}(t) - \boldsymbol{y}(t). \tag{8}$$

In a PID controller, the control variable $\boldsymbol{u}(t)$ is composed of the proportional (P), integral (I), and derivative (D) terms and given by:

$$\boldsymbol{u}(t) = K_p \boldsymbol{e}(t) + K_i \int_0^t \boldsymbol{e}(\tau) d\tau + K_d \frac{d\boldsymbol{e}(t)}{dt}, \tag{9}$$

where $K_p, K_i, K_d \geq 0$ are the proportional, integral, and derivation gains, respectively. In PID control design, the P, I, and D play different roles: *Proportional term (P)* outputs a correction proportional to the current error $\boldsymbol{e}_t$, but alone leaves a steady-state offset; *Integral term (I)* accumulates past errors to remove residual bias, ensuring offsets are corrected even as proportional effects fade; and *Derivative term (D)* responds to the error's rate of change, damping rapid growth to improve stability and reduce overshoot.

State-Feedback PID Controller. A special case of the PID controller is obtained by choosing the measured output $\boldsymbol{y}(t)$ to be the state variable $\boldsymbol{x}(t)$ in Eqn. 7, yielding the following state-space model

$$\dot{\boldsymbol{x}}(t) = g(\boldsymbol{x}(t),\boldsymbol{u}(t),t), \boldsymbol{y}(t) = \boldsymbol{x}(t). \tag{10}$$

The control error then becomes the state tracking error, $\boldsymbol{e}(t) = \boldsymbol{x}_{sp}(t) - \boldsymbol{x}(t)$, and the system is controlled through feedback of the state (Åström & Murray, 2021).

3 Steering with a Feedback Controller

In this section, we will formulate popular activation steering methods, such as ActAdd, DirAblate, and Mean-AcT, as a state-feedback P controller. Based on this new interpretation, we propose PID Steering, a novel steering method that uses a PID controller.

3.1 Activation Steering as a P Controller

We consider the state-feedback PID controller given in Eqn. 10 and the continuous steering vector $\boldsymbol{r}(t)$ in which we replace the layer index k by the time index t. Substituting the state tracking error $\boldsymbol{e}(t)$ by the difference-in-means vector $\boldsymbol{r}(t)$ and using the P controller whose system dynamics is governed by $g(\boldsymbol{x}(t),\boldsymbol{u}(t),t) = f(\rho_{\text{steer}}(\boldsymbol{x}(t),\boldsymbol{u}(t)),t) - \boldsymbol{x}(t)$, we obtain

$$\dot{\boldsymbol{x}}(t) = f(\rho_{\text{steer}}(\boldsymbol{x}(t),K_p\boldsymbol{r}(t)),t) - \boldsymbol{x}(t). \tag{11}$$

We discretize Eqn. 11 using Euler method (Euler, 1768; Hairer et al., 1993) to obtain

$$\boldsymbol{x}(k)-\boldsymbol{x}(k-1)=f^{(k)}(\rho_{\text{steer}}(\boldsymbol{x}(k-1),K_p\boldsymbol{r}(k-1)))-\boldsymbol{x}(k-1),$$

or equivalently,

$$\boldsymbol{x}(k)=f^{(k)}(\rho_{\text{steer}}(\boldsymbol{x}(k-1),K_p\boldsymbol{r}(k-1))), \tag{12}$$

where $f^{(k)}(\cdot)=f(\cdot,k)$, a function depending on index k.

Comparing Eqn. 12 with Eqn. 1 and 2 shows that applying the steering vectors as in Section 2.2.1 is equivalent to implementing the P controller, where $f^{(k)}$ is the k-th layer in an LLM, $\boldsymbol{u}(t)=K_p\boldsymbol{r}(t)$ is the new steering vector. Thus, activation steering computes the expected state tracking error.

$$\boldsymbol{r}(t)=\bar{\boldsymbol{e}}(t)=\mathbb{E}_{\boldsymbol{q}_{sp}\in\mathcal{D}^{(\text{train})}_{\text{target}}}[\boldsymbol{x}_{sp}(t,\boldsymbol{q}_{sp})]-\mathbb{E}_{\boldsymbol{q}\in\mathcal{D}^{(\text{train})}_{\text{source}}}[\boldsymbol{x}(t,\boldsymbol{q})]. \tag{13}$$

This expected state tracking error, i.e., the difference-in-means vector $\boldsymbol{r}(t)$, can be computed non-sequentially or sequentially, as explained in Section 2.2.2. When $\boldsymbol{r}(t)$ is computed non-sequentially and $\rho_{\text{steer}}(\boldsymbol{x}(k),\boldsymbol{u}(k))=\boldsymbol{x}(k)+\alpha\boldsymbol{u}(k)$ or $\boldsymbol{x}(k)-\boldsymbol{u}(k)\,\boldsymbol{u}(k)^\top\,\boldsymbol{x}(k)$, we obtain ActAdd or DirAblate, respectively. When $\boldsymbol{r}(t)$ is computed sequentially and $\rho_{\text{steer}}(\boldsymbol{x}(k),\boldsymbol{u}(k))=\boldsymbol{x}(k)+\alpha\boldsymbol{u}(k)$, we attain Mean-AcT.

Limitations of P Controller. There is always a steady state error in P control. The error decreases with increasing gain, but the tendency towards oscillation also increases. Since activation steering methods, i.e., ActAdd, DirAblate, and Mean-Act, are P controllers, they share the same limitations. We informally state our theoretical guarantees that P-control activation steering methods cannot alleviate the steady state error in Proposition 1 below and provide detailed proofs in Appendix B.4.

Proposition 1 (Steady-state error of P-control activation steering) *P-control activation steering ensures input-to-state stability (ISS) for an appropriate range of K_p. However, there still exists a steady-state error due to the disturbance $\boldsymbol{w}(k)$ to the state of the system. In the best case, when $\boldsymbol{w}(k)$ converges to $\boldsymbol{w}$, under a mild condition, the expected error, i.e., the difference-in-means, $\boldsymbol{r}(k)=\bar{\boldsymbol{e}}(k)$ eventually converges to a steady state $\bar{\boldsymbol{e}}_{ss}\propto\boldsymbol{w}$. Therefore, $\bar{\boldsymbol{e}}_{ss}\neq 0$ if $\boldsymbol{w}\neq 0$.*

We further provide empirical evidence to validate Proposition 1 in Figure 3 below. To archive this, we apply Sequential P-control activation steering (P Steering) on a randomly intitialized model with 150 layers deep, and pretrained Qwen2.5-3B-Instruct. We use $\langle\bar{e}(0),\bar{e}(t)\rangle$ as metric since it is a scalar measure of the energy retained along $\bar{e}(0)$. If this quantity fails to decay to zero, e.g., under noise, it indicates a persistent component of the initial error, i.e., an undesired dynamic (see Appendix B.6 for further explanation). It can be seen that, for both the randomly initialized and the pretrained models, the errors do not vanish completely. These results confirm that P-control activation steering ensures stability but admits a persistent steady-state error due to the disturbance.

(a) Randomly Initialized LLama3 (b) Pretrained Qwen2.5-3B-Instruct

Figure 3: Scalar errors across time step of randomly initialized model after applying P, PI, and PID controller.

3.2 Proportional–Integral–Derivative (PID) Steering

3.2.1 Overview

To overcome the steady-state error inherent in P-control activation steering, we extend the method by adding integral (I) and derivative (D) terms to the steering vectors. PID Steering thus (i) reacts immediately to errors via the P term for greater responsiveness, (ii) removes steady-state offsets with the I term, ensuring convergence to the desired set point, and (iii) anticipates error trends through the D term, improving stability and reducing overshoot. Together, these properties yield the following advantages:

- **Generalization.** PID Steering extends P-control methods like ActAdd, DirAblate, and Mean-AcT by adding integral and derivative components.

- **Methodological Agnosticism.** Our PID framework can be applied across different activation steering techniques, including ActAdd, DirAblate, and Mean-AcT.
- **Stability.** We theoretical prove and empirical demonstrate that PID Steering reduces steady-state error and overshoot in P-controllers, improving existing steering methods.
- **Interpretability.** Derived from classical feedback control (Minorsky, 1922), the framework inherits the simplicity and interpretability that underpin the wide use of PID controllers.

3.2.2 Computing the Steering Direction Using a PID Feedback Controller

Following Section 3.1, we consider the state-feedback PID controller in Eqn. 10, replacing the state tracking error $\boldsymbol{e}(t)$ with the difference-in-means vector $\boldsymbol{r}(t)$. With the PID controller governed by the system dynamics $g(\boldsymbol{x}(t),\boldsymbol{u}(t),t)=f(\rho_{\text{steer}}(\boldsymbol{x}(t),\boldsymbol{u}(t)),t)-\boldsymbol{x}(t)$, we obtain

$$\dot{\boldsymbol{x}}(t)=f(\rho_{\text{steer}}(\boldsymbol{x}(t),K_p\boldsymbol{r}(t)+K_i\int_0^t\boldsymbol{r}(\tau)d\tau+K_d\frac{d\boldsymbol{r}(t)}{dt}),t)-\boldsymbol{x}(t). \tag{14}$$

Eqn. 14 defines the continuous-time model of PID Steering, whose steering vector is given by:

$$\boldsymbol{u}(t)=K_p\boldsymbol{r}(t)+K_i\int_0^t\boldsymbol{r}(\tau)d\tau+K_d\frac{d\boldsymbol{r}(t)}{dt}. \tag{15}$$

In order to obtain the discrete-time formulation of PID Steering, we first discretize the sytem dynamics $\dot{\boldsymbol{x}}(t)=f(\rho_{\text{steer}}(\boldsymbol{x}(t),\boldsymbol{u}(t)),t)-\boldsymbol{x}(t)$ using Euler method (Euler, 1768; Hairer et al., 1993), same as in Section 3.1, and attain

$$\boldsymbol{x}(k)=f^{(k)}(\rho_{\text{steer}}(\boldsymbol{x}(k-1),\boldsymbol{u}(k-1))), \tag{16}$$

Next, we discretize $\boldsymbol{u}(t)$ given in Eqn. 15 to obtain $\boldsymbol{u}(k)$ using Lemma 1 below.

Lemma 1 (Discretizing PID steering vector) *Consider the continuous PID steering vector defined in Eqn. 15. The discrete-time PID steering vector is given by:*

$$\boldsymbol{u}(k)=K_p\boldsymbol{r}(k)+K_i\sum_{j=0}^{k-1}\boldsymbol{r}(j)+K_d(\boldsymbol{r}(k)-\boldsymbol{r}(k-1)). \tag{17}$$

Proof of Lemma 1 is in Appendix B.1. With Eqn. 16 and Lemma 1, we now define PID Steering.

Definition 1 (PID Steering) *Given a large language model whose layers are $\{f^{(k)}\}_{k=1}^{K}$ and a steering function ρ_{steer}, PID Steering constructs the steering vectors as follows:*

$$\boldsymbol{u}(k)=K_p\boldsymbol{r}(k)+K_i\sum_{j=0}^{k-1}\boldsymbol{r}(j)+K_d(\boldsymbol{r}(k)-\boldsymbol{r}(k-1)), \tag{18}$$

where for non-sequential mapping,

$$\boldsymbol{r}(k)=\mathbb{E}_{\boldsymbol{q}_{sp}\in\mathcal{D}_{target}^{(train)}}[\boldsymbol{x}_{sp}(k,\boldsymbol{q}_{sp})]-\mathbb{E}_{\boldsymbol{q}\in\mathcal{D}_{source}^{(train)}}[\boldsymbol{x}(k,\boldsymbol{q})],$$

and for sequential mapping,

$$\tilde{\boldsymbol{x}}(k)=f^{(k)}\big(\rho_{steer}\big(\boldsymbol{x}(k-1),\boldsymbol{u}(k-1)\big)\big),\quad \boldsymbol{r}(k)=\mathbb{E}_{\boldsymbol{q}_{sp}\in\mathcal{D}_{target}^{(train)}}[\boldsymbol{x}_{sp}(k,\boldsymbol{q}_{sp})]-\mathbb{E}_{\boldsymbol{q}\in\mathcal{D}_{source}^{(train)}}[\tilde{\boldsymbol{x}}(k,\boldsymbol{q})].$$

4 Theoretical Analysis of PID Steering

This section provides theoretical evidence for our claims: (i) adding integral action (PI) reduces steady-state error that remains under pure P-control (Proposition 3); and (ii) adding a derivative term (PID) preserves bias removal while mitigating oscillations/overshoot (Proposition 1 and 2). We denote $\boldsymbol{K}_p:=K_p\boldsymbol{I}$, $\boldsymbol{K}_i:=K_i\boldsymbol{I}$, and $\boldsymbol{K}_d:=K_d\boldsymbol{I}$. Detailed proofs are provided in Appendix B.

4.1 Dynamics of the Average Error Across Layers

To formalize the problem, we consider N pairs of prompts/input tokens from two contrastive datasets, e.g. harmful and harmless, $\{(\boldsymbol{q}_i^+,\boldsymbol{q}_i^-)\}_{i=1}^N$ with corresponding activations $\boldsymbol{x}_i^{\pm}(k)\in\mathbb{R}^d$ at layer k. A steering input $\boldsymbol{u}(k)$ perturbs the undesired branch:

$$\boldsymbol{x}_i^-(k+1)=f_i^{(k)}\big(\boldsymbol{x}_i^-(k)+\boldsymbol{u}(k)\big). \tag{19}$$

Let $\bar{\boldsymbol{e}}(k):=\frac{1}{N}\sum_{i=1}^N(\boldsymbol{x}_i^+(k)-\boldsymbol{x}_i^-(k))$, the error dynamics of activation steering is then given by Proposition 2 below.